Scientia Generalis 2675-2999

v. 7, n. 2, p. 105-113. 2026.

Artigo Original

LARVICIDAL ACTIVITY OF THE β-CARBOLINE ALKALOID HARMINE ISOLATED FROM *BANISTERIOPSIS CAAPI* AGAINST *AEDES AEGYPTI*

DOI: 10.22289/sg.V7N2A10

Fernanda Dias Viana [a*], Andrei da Silva Alexandre [a], Rosemary Aparecida Roque [b], Ana Cristina da Silva Pinto [b] e Waldireny Rocha Gomes [a]

[a] *Universidade Federal do Amazonas - UFAM, Manaus, Amazonas, Brazil.*
[b] *Instituto Nacional de Pesquisas da Amazônia - INPA, Manaus, Amazonas, Brazil.*

Abstract

The mosquito *Aedes aegypti* is the main vector of dengue, Zika, and Chikungunya, diseases that affect millions of people annually in the Americas. Control of this vector relies mainly on the use of synthetic insecticides, which present limitations related to resistance development and environmental impacts. In this context, β-carboline alkaloids present in *Banisteriopsis caapi* have attracted interest due to their neurotoxic activity in insects, suggesting potential larvicidal applications. This study aimed to isolate harmine from *B. caapi* leaves and evaluate its larvicidal activity against *A. aegypti*. Dried leaves were subjected to sequential maceration with hexane, ethyl acetate, and ethanol. Harmine purification was carried out using column chromatography and preparative chromatography. Structural identification was confirmed by high-resolution mass spectrometry (LC-HRMS/MS) and proton and carbon-13 nuclear magnetic resonance spectroscopy (^{1}H and ^{13}C NMR). Larvicidal bioassays were conducted using third-instar *A. aegypti* larvae following a protocol adapted from the World Health Organization (WHO). The crude extracts showed mortality rates of 3% for the hexane extract, 7% for the ethyl acetate extract, and 67% for the ethanolic extract. Isolated harmine showed a mortality rate of 40%. These results demonstrate the moderate potential of harmine as a natural larvicide against *A. aegypti*, providing a basis for further investigations and for the development of bioinsecticides derived from Amazonian resources.

Keywords: Mosquito; Botanical insecticide; Vector control; Larvicidal bioassay; Extract.

ATIVIDADE LARVICIDA DO ALCALOIDE β-CARBOLÍNICO HARMINA ISOLADO DE BANISTERIOPSIS CAAPI FRENTE AO AEDES AEGYPTI

Resumo

O mosquito *Aedes aegypti* é o principal vetor da dengue, Zika e Chikungunya, doenças que afetam milhões de pessoas anualmente nas Américas. O controle desse vetor se baseia principalmente no uso de inseticidas sintéticos, os quais apresentam limitações relacionadas ao desenvolvimento de resistência e aos impactos ambientais. Nesse contexto, os alcaloides β-carbolínicos presentes em *Banisteriopsis caapi* têm despertado interesse devido à sua atividade neurotóxica em insetos, sugerindo potencial aplicação como agentes larvicidas. Este estudo teve como objetivo isolar a harmina das folhas de *B. caapi* e avaliar sua atividade larvicida contra *A. aegypti*. Folhas secas foram submetidas à maceração sequencial com hexano, acetato de etila e etanol. A purificação da harmina foi realizada por cromatografia em coluna e cromatografia preparativa. A identificação estrutural foi confirmada por espectrometria de massas de alta resolução (LC-HRMS/MS) e espectroscopia de ressonância magnética nuclear de hidrogênio e carbono-13 (RMN de ^{1}H e ^{13}C). Os bioensaios larvicidas foram conduzidos utilizando larvas de terceiro instar de *A. aegypti*, seguindo um protocolo adaptado da Organização Mundial da Saúde (OMS). Os extratos brutos apresentaram taxas de mortalidade de 3% para o extrato hexânico, 7% para o extrato em acetato de etila e 67% para o extrato etanólico. A harmina isolada apresentou uma taxa de mortalidade de 40%. Esses resultados demonstram o potencial moderado da harmina como larvicida natural contra *A. aegypti*, fornecendo uma base para futuras investigações e para o desenvolvimento de bioinseticidas derivados de recursos amazônicos.

Palavras-chave: Mosquito; Inseticida botânico; Controle vetorial; Bioensaio larvicida; Extrato.

[*] Autor para correspondência: nandafediasviana@gmail.com

Artigo recebido em 25/07/2026 e aprovado para publicação em 13/08/2026.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

ACTIVIDAD LARVICIDA DE LA ALCALOMINA B-CARBOLINA AISLADA DE *BANISTERIOPSIS CAAPI* FRENTE A *AEDES AEGYPTI*

Resumen

El mosquito *Aedes aegypti* es el principal vector del dengue, el Zika y el Chikungunya, enfermedades que afectan a millones de personas cada año en América. El control de este vector se basa principalmente en el uso de insecticidas sintéticos, que presentan limitaciones relacionadas con el desarrollo de resistencia y los impactos medioambientales. En este contexto, los alcaloides β-carbolinos presentes en *Banisteriopsis caapi* han despertado interés debido a su actividad neurotóxica en insectos, lo que sugiere posibles aplicaciones larvicidales. Este estudio tenía como objetivo aislar la harmina de las hojas de *B. caapi* y evaluar su actividad larvicida frente a *A. aegypti*. Las hojas secas fueron sometidas a maceración secuencial con hexano, acetato de etilo y etanol. La purificación de la harmina se realizó mediante cromatografía en columna y cromatografía preparativa. La identificación estructural fue confirmada mediante espectrometría de masas de alta resolución (LC-HRMS/MS) y espectroscopía de resonancia magnética nuclear de protones y carbono-13 (RMN ^{1}H y ^{13}C). Los bioensayos larvicidas se realizaron utilizando larvas de tercer estadio de *A. aegypti* siguiendo un protocolo adaptado de la Organización Mundial de la Salud (OMS). Los extractos crudos mostraron tasas de mortalidad del 3% para el extracto de hexano, 7% para el extracto de acetato de etilo y 67% para el extracto etanol. La harmina aislada mostró una tasa de mortalidad del 40%. Estos resultados demuestran el potencial moderado de la harmina como larvicida natural contra *A. aegypti*, proporcionando una base para investigaciones posteriores y para el desarrollo de bioinsecticidas derivados de recursos amazónicos.

Palabras clave: Mosquito; Insecticida botánico; Control vectorial; Bioensayo larvicida; Extracto.

1. Introduction

Aedes aegypti (Diptera: Culicidae) is recognized as the principal vector of arboviruses in the Americas, transmitting dengue, Zika, and chikungunya (WHO, 2017). In 2023 alone, the Region of the Americas reported 4,569,464 dengue cases, including 7,665 severe cases and 2,363 deaths (PAHO, 2024). In the state of Amazonas, 8,161 probable cases were documented in 2024, and by Epidemiological Week 4 of 2025, Brazil had already registered 160,984 cases and 33 deaths (Brazil, 2025).

Vector control strategies rely predominantly on organophosphates and pyrethroids; however, resistant populations of *A. aegypti* have been identified in several Brazilian states (Moyes *et al*., 2017). Moreover, these chemical compounds are associated with toxicity to non-target organisms and extended environmental persistence. Plant-derived alkaloids represent a promising alternative, given their faster degradation rates and reduced selective pressure for resistance development (Isman, 2006).

Banisteriopsis caapi (Spruce ex Griseb.) C.V. Morton (Malpighiaceae), an Amazonian liana, is widely recognized as a key ingredient in the psychoactive brew ayahuasca. Its pharmacological properties include bactericidal, fungicidal, and antiviral activities, largely attributed to the β-carboline alkaloids harmine, harmaline, and tetrahydroharmine (Cao *et al*., 2007; Santos *et al*., 2020). These compounds act as reversible inhibitors of monoamine oxidase A (MAO-A), thereby modulating neurotransmission in both vertebrates and invertebrates (Passos *et al*., 2014).

Noorazlan *et al*. (2024) demonstrated the larvicidal potential of *B. caapi* extracts against *Aedes albopictus*, suggesting possible efficacy against *A. aegypti*. Nevertheless, no study has specifically investigated the activity of isolated harmine. The present work therefore aimed to extract, purify, and identify harmine from *B. caapi* leaves and to evaluate its larvicidal activity against *A. aegypti*.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

2. Materials and methods

2.1 Plant Material and Extract Preparation

Leaves of *Banisteriopsis caapi* were collected in December 2024 at the Federal University of Amazonas (UFAM), southern sector, Manaus-AM (SISGEN authorization: A867BF8), and taxonomically identified by botanist Dr. Ari de Freitas Hidalgo. The plant material was air-dried at room temperature and subsequently ground to a uniform particle size. For sequential maceration, 210.79 g of leaves were extracted successively with hexane, ethyl acetate, and ethanol (3 × 1 L of each solvent, 24 h per cycle). The resulting extracts were filtered and concentrated using a rotary evaporator at 45°C under reduced pressure.

The infusion and decoction procedures were done with 200 g of dried *B. caapi* leaves, using 100 g of material for each 800 mL of distilled water. The aqueous extracts obtained were subsequently lyophilized at the Analytical Center Laboratory of UFAM.

2.2 Purification by Chromatography

The ethanolic extract (5.42 g) was subjected to column chromatography (silica gel 60, 0.063–0.200 mm, 40 × 3 cm column), employing a gradient elution system of increasing polarity: 100% hexane, hexane/dichloromethane (gradient), dichloromethane/methanol (gradient), and finally 100% methanol. A total of 84 fractions were collected and monitored by thin-layer chromatography (TLC, silica gel F254), with visualization under UV light at 254 and 365 nm, followed by Dragendorff's reagent.

Fractions exhibiting similar chromatographic profiles were reunited. Fraction 81 was further purified by preparative chromatography on glass plates (20 × 20 cm, silica gel G, 1 mm thickness), eluted with chloroform/methanol (9:1, v/v), following the methodology described by Jiang *et al*. (2023). Bands were visualized under UV light (254 and 365 nm), scraped, and re-extracted with the same solvent system. The final yield of isolated harmine was 33 mg, with an observed Rf value of 0.3.

2.3 Structural Identification

2.3.1 Mass Spectrometry

Analyses were conducted using a high-resolution ion trap mass spectrometer, model LCQ Fleet (Thermo Scientific®), equipped with an electrospray ionization (ESI) source operating in positive mode. The instrumental parameters were as follows: charging current of 5 µA, vaporizer temperature of 320 °C, capillary temperature of 220 °C, sheath gas pressure of 30 psi, auxiliary gas flow of 10 arb, and a mass detection range of *m/z* 100–1000. Tandem mass spectrometry (MS/MS) experiments were performed employing helium as the collision gas, with collision energies ranging from 20–30% to enable structural confirmation. Data acquisition and processing were carried out using Xcalibur software.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

2.3.2 Nuclear Magnetic Resonance

^{1}H and ^{13}C NMR spectra of the isolated compounds were acquired using a monodimensional nuclear magnetic resonance spectrometer. Measurements were performed on a Bruker Avance III HD instrument operating at 500 MHz for ^{1}H and 125 MHz for ^{13}C, with $CDCl_3$ as the solvent and tetramethylsilane (TMS) as the internal reference. Chemical shifts (δ) were expressed in ppm and referenced to the residual solvent signal ($CDCl_3$: δH 7.26; δC 77.16). All analyses were conducted at the Nuclear Magnetic Resonance Laboratory (NMRLAB) of the Analytical Center, Federal University of Amazonas (UFAM). Spectral data were processed using Mestre Nova software version 2.0.7, and molecular structures were generated with ACD/Chemsketch software version 22.2.3 (ACDlabs).

2.4 Larvicidal Bioassays

A. aegypti larvae (third instar) were obtained from the Malaria and Dengue Vector Laboratory of the National Institute for Amazonian Research (INPA). Bioassays were conducted following a protocol adapted from the World Health Organization (WHO, 2005). All experiments were performed in a climate-controlled environment maintained at 25 ± 2°C and 72 ± 5% relative humidity.

For preliminary assays with crude extracts, samples (20 mg) were dissolved in 1% dimethyl sulfoxide (DMSO). Each sample solution was applied to test cups containing 9.6 mL of distilled water and 10 *Aedes* larvae, resulting in a final concentration of 500 µg/mL. The negative control consisted of distilled water with 1% DMSO only. Larval mortality was assessed after 24 h.

Median lethal concentration (LC_{50}) values were determined by log-probit regression analysis using Polo Plus 1.0 software (LeOra Software, CA, USA), with 95% confidence intervals. Assays were considered valid when mortality in the control group was below 10%.

To determine the LC_{50} (lethal concentration required to kill 50% of the larval population), additional bioassays were performed using ethanolic extract concentrations that produced mortality rates above 30% in the preliminary screening. Serial dilutions of the *B. caapi* ethanolic extract were prepared at concentrations of 25, 50, 100, 250, 500, and 1,000 µg/mL. Based on Probit analysis, the LC_{50} value of the ethanolic extract was estimated to be greater than 1,000 µg/mL, indicating moderate larvicidal activity under the experimental conditions evaluated.

3. Results and discussion

3.1 Extract Yield and Purification

Sequential maceration of dried *Banisteriopsis caapi* leaves yielded extraction efficiencies of 2.56% with hexane, 8.29% with ethyl acetate, and 5.08% with ethanol. Decoction and infusion each yielded 8 g of extract, corresponding to an extraction yield of 8%. Overall, five distinct extracts were obtained using different extraction methodologies.

Column chromatography followed by preparative chromatography enabled the isolation of harmine, with a final mass of 33 mg. The observed Rf value of 0.3 in the chloroform/methanol system (9:1, v/v) is consistent with that reported by Jiang *et al*. (2023) for harmine.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

3.2 Structural Identification of Harmine

3.2.1 Mass Spectrometry

ESI(+)-HRMS analysis confirmed the identity of harmine. The spectrum exhibited a protonated molecular ion $[M+H]^+$ at *m/z* 213 (calculated for $C_{13}H_{12}N_2O$: 213.1022). The principal MS/MS fragment ions included *m/z* 198, corresponding to the loss of a methyl group (15 Da), *m/z* 170, attributed either to the combined loss of CH_3 and CO or to fragmentation of the β-carboline nucleus, *m/z* 157, indicative of deeper fragmentation within the aromatic system, and *m/z* 196 corresponding to the loss of ammonia (NH_3, 17 Da) from the protonated molecule (Liu, 2013; Huber, 2025).

3.2.2 Nuclear Magnetic Resonance

The ^1H and ^{13}C NMR spectra confirmed the structure of harmine. The ^1H NMR data are reported as δ (ppm), multiplicity, coupling constant *J* (Hz), integration, and assignment: δ 2.86 (s, 3H, 1-CH_3); δ 3.91 (s, 3H, O-CH_3); δ 6.90 (dd, *J* = 2.1, 8.6 Hz, 1H, H-6); δ 7.02 (s, 1H, H-8); δ 7.66 (d, *J* = 5.3 Hz, 1H, H-4); δ 7.91 (d, *J* = 8.6 Hz, 1H, H-5); δ 8.17 (d, *J* = 5.3 Hz, 1H, H-3); δ 9.38 (s, 1H, N-H).

The ^{13}C NMR data are as follows: δ 14.13 (C-1), δ 135.83 (C-3), δ 112.33 (C-4), δ 122.78 (C-5), δ 110.72 (C-6), δ 161.50 (C-7), δ 94.79 (C-8), δ 134.39 (C-10), δ 115.22 (C-11), δ 129.67 (C-12), δ 142.87 (C-13), δ 22.70 (CH_3-1), and δ 55.67 (OCH_3-7).

The chemical shift values obtained are consistent with those reported in the literature for harmine (Berrougui, 2006), thereby providing unequivocal confirmation of the structure of the isolated alkaloid.

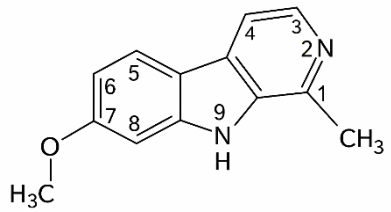


Figure 1. Chemical structure of harmine (1). Source: Research data (2026).

3.3 Larvicidal Activity

The five crude extracts obtained through different extraction methodologies, together with the purified compound harmine, exhibited distinct larvicidal activity and mortality rates after 24 h at the tested concentration (Table 1). The higher mortality observed in the ethanolic extract suggests a greater abundance of active polar constituents, notably β-carboline alkaloids. Consequently, subsequent dilution assays were performed exclusively with the ethanolic extract, as it was the only sample to induce mortality exceeding 40%, thereby permitting the calculation of LC_{50} values.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

Table 1. Percentage of larvicidal mortality of crude extracts and harmine.

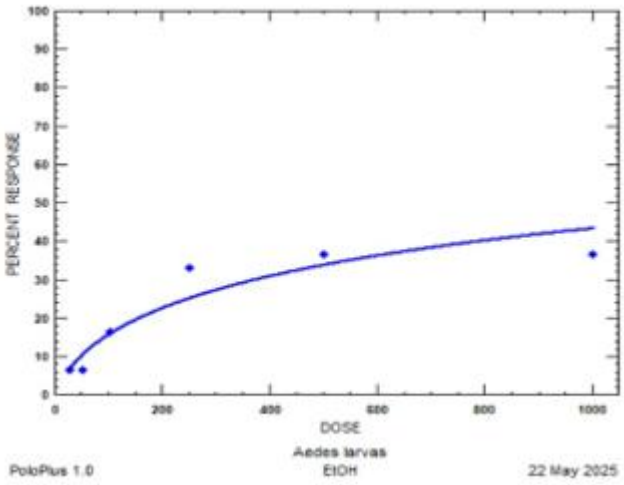
Sample	Concentration (µg/mL)	% of larvicidal mortality
Hexane extract	500	3
Ethyl acetate extract	500	7
Ethanolic	500	67
Infusion extract	500	0
Decoction extract	500	0
Harmine	500	40

Table 2. LC_{50} of the ethanolic extract of *Banisteriopsis caapi* against *Aedes aegypti* larvae.

Parameter	**Value**
LC_{50} (µg/mL)	1000
Chi- square	2,192
Degree of freedom	4

Source: Research data (2026).

Figure 2. Probit regression plot of the larvicidal activity of the ethanolic extract against *A. aegypti* larvae.



Source: Research data (2026).

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

The ethanolic extract of *B. caapi* exhibited greater larvicidal activity against *A. aegypti* than isolated harmine under the experimental conditions employed. The purified alkaloid produced only moderate larval mortality, whereas the crude extract elicited a stronger biological response, indicating that the larvicidal effect of *B. caapi* cannot be attributed to harmine alone. These results suggest that multiple constituents act in combination to produce the activity observed.

This finding is consistent with the phytochemical profile of *B. caapi*. Its ethanolic extracts contain a complex mixture of *β*-carboline alkaloids, including harmine, harmaline, and tetrahydroharmine, together with phenolic compounds and other secondary metabolites. Because ethanol extracts compounds across a broad range of polarities, it preserves much of the chemical complexity present in the plant. The greater activity of the crude extract may therefore result from additive or synergistic interactions among these constituents, rather than from the effect of a single metabolite. This pattern has been documented outside the *Aedes* system as well, in stored-grain coleopterans, Nenaah (2011) evaluated methanol extract, isolated *β*-carboline alkaloids, and binary alkaloid mixtures from *Peganum harmala* seeds against *Rhyzopertha dominica* and *Tribolium castaneum*. Jiang *et al*. (2023) reported a comparable pattern in a mosquito system: an equimolar mixture of harmine and harmaline showed greater larvicidal activity against *Aedes albopictus* than either alkaloid tested alone, supporting the hypothesis that interactions among *β*-carbolines influence biological activity.

Although harmine displayed only moderate larvicidal activity in the present study, its isolation confirms that one of the principal *β*-carboline alkaloids of *B. caapi* possesses intrinsic activity against *A. aegypti*. This is consistent with reports of harmine's insecticidal potential in other insect systems: Zeng *et al*. (2010) showed that harmine and synthetic *β*-carboline derivatives had insecticidal activity against fourth-instar larvae of *Culex pipiens quinquefasciatus*, and Bouayad *et al*. (2012) found that dietary harmine reduced growth, increased mortality, and inhibited *α*-amylase activity in larvae of the stored-product moth *Plodia interpunctella*. These studies used different insect taxa and exposure routes (dietary ingestion rather than aquatic immersion), so the comparison is illustrative of harmine's general bioactivity rather than a direct equivalence to the present bioassay. The higher efficacy of the crude *B. caapi* extract relative to isolated harmine, however, indicates that harmine is not the sole determinant of larval mortality in the present system. Minor alkaloids or other classes of secondary metabolites may contribute directly to the activity observed, or may modulate the effects of harmine through complementary mechanisms. This underscores the value of evaluating complex extracts alongside isolated constituents when investigating plant-derived insecticides.

The larvicidal activity observed in the present study was lower than that reported by Jiang *et al*. (2023), who found LC_{50} values of 44.54 ± 2.56 µg/mL for the total alkaloid extract, 55.51 ± 3.01 µg/mL for harmaline, 93.67 ± 4.53 µg/mL for harmine, and 117.87 ± 5.61 µg/mL for harmalol against *A. albopictus*. Direct comparisons between studies require caution, as mosquito species, colony origin, extraction protocol, phytochemical composition of the tested material, and bioassay conditions vary considerably and may substantially affect larval susceptibility.

The mechanism underlying the larvicidal activity of *β*-carboline alkaloids was not investigated in the present study. *β*-carboline alkaloids are well established inhibitors of monoamine oxidase (MAO) in mammalian systems (Abrantes, 2022). Whether this mechanism extends to insects is less clear, insects lack a well-characterized MAO homolog, and octopaminergic signaling, which regulates locomotion, feeding, muscle contraction, and sensory processing in insects, is more commonly implicated as a direct target of insecticidal phytochemicals, independent of MAO-type inhibition. The results reported here are therefore compatible with a disruption of monoaminergic signaling in the larvae, but should not be presented as evidence of MAO inhibition specifically, since the enzymatic basis for that mechanism in *A. aegypti* has not been established.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

The moderate activity of the ethanolic extract does not, on its own, support immediate practical use as a larvicide. It does, however, identify *B. caapi* as a source of bioactive metabolites warranting further investigation, since crude extracts commonly serve as the starting point for identifying active chemical combinations or optimizing fractions with improved biological performance. These findings provide a basis for future bioassay-guided fractionation, quantitative phytochemical characterization, and evaluation of interactions among the principal *β*-carboline alkaloids.

Certain limitations should be considered when interpreting these findings. Bioassays were conducted exclusively under laboratory conditions, using third-instar *A. aegypti* larvae and a single 24-hour exposure period. The chemical composition of the ethanolic extract was not quantitatively determined, which precludes identification of the metabolites that contributed most strongly to the activity observed; only harmine was structurally characterized. The proposed mechanism of action was likewise inferred from previous studies rather than demonstrated experimentally. Future investigations should include quantitative phytochemical analysis, bioassay-guided fractionation, and enzymatic and molecular approaches to clarify the mode of action, together with toxicity testing in non-target aquatic organisms and semi-field or field evaluations.

4. Conclusion

Among the six extracts obtained through different extraction methodologies, the ethanolic extract exhibited the highest larvicidal activity against *A. aegypti*. From this extract, harmine was successfully isolated, and its structure was confirmed by high-resolution mass spectrometry and nuclear magnetic resonance analyses. The alkaloid demonstrated larvicidal activity, inducing 40% mortality in *A. aegypti* larvae.

Further investigations are required to elucidate the precise mechanism of action, determine potential toxicity to non-target organisms, and evaluate the feasibility of developing formulations suitable for field application in vector control programs. This study highlights the relevance of Amazonian biodiversity as a source of bioactive compounds and contributes to the advancement of sustainable bioinsecticides.

5. Acknowledgements

The authors express their gratitude to the Federal University of Amazonas (UFAM) for providing the laboratory infrastructure, and to the Nuclear Magnetic Resonance Laboratory (NMRLAB) at the Analytical Center of UFAM for the mass spectrometry and NMR services. Special thanks are extended to the Malaria and Dengue Laboratory of the National Institute of Amazonian Research (INPA) for supplying *Aedes aegypti* larvae. Financial support was provided by the Amazonas State Research Support Foundation (FAPEAM) through a master's scholarship granted under the Postgraduate Program in Pharmaceutical Sciences at UFAM.

References

ABRANTES, T. G. de L. B. **Health risks of ayahuasca users: an integrative review**. Mossoró, RN, 2022.

ARAÚJO, G. N. *et al*. **Extraction and analysis of alkaloids in Banisteriopsis caapi leaves**. Academic Integration Symposium "Basic Sciences for Sustainable Development" SAI UFV, 2023.

Scientia Generalis, v. 7, n. 2, p. 105-113. 2026.

BERROUGUI, H.; MARTINEZ, M.; OUELLETTE, M.; ISMAILI, M.; KHALIL, A.; MANSOURI, A.; HADDAD, P. S. Vasorelaxant effects of harmine and harmaline extracted from Peganum harmala L. seeds in isolated rat aorta. **Pharmacological Research**, v. 53, n. 2, p. 170-177, 2006. doi: 10.1016/j.phrs.2005.10.005.

BOUAYAD, N. *et al*. Dietary effects of harmine, a β-carboline alkaloid, on development, energy reserves and α-amylase activity of *Plodia interpunctella* Hübner (Lepidoptera: Pyralidae). **Saudi Journal of Biological Sciences**, v. 19, n. 1, p. 73-80, 2012. doi: 10.1016/j.sjbs.2010.12.004.

BRASIL. Ministério da Saúde. Secretaria de Vigilância em Saúde e Ambiente. Departamento de Doenças Transmissíveis. **Plano de contingência nacional para dengue, chikungunya e zika**. Brasília, DF, 2025.

BRAZIL. **Ministry of Health.** *Aedes aegypti*. 2025. Disponível em: https://www.gov.br/saude/pt-br/assuntos/saude-de-a-a-z/a/aedes-aegypti. Acesso em: 15 abr. 2026.

CAO, R. *et al*. β-Carboline alkaloids: Biochemical and pharmacological functions. **Current Medicinal Chemistry**, v. 14, n. 4, p. 479–500, 2007. doi: 10.2174/092986707779940998.

HUBER, B.; *et al*. Metabolic profiling reveals first evidence of fumigation drug plant Peganum harmala in Iron Age Arabia. **Communications Biology**, v. 8, n. 1, p. 1-12, 2025. doi: 10.1038/s42003-025-02234-6.

ISMAN, M. B. Botanical insecticides, deterrents, and repellents in modern agriculture and an increasingly regulated world. **Annual Review of Entomology**, v. 51, p. 45-66, 2006. doi: 10.1146/annurev.ento.51.110104.151146.

JIANG, N. *et al*. Lethal and sublethal toxicity of beta-carboline alkaloids from *Peganum harmala* (L.) against *Aedes albopictus* larvae (Diptera: Culicidae). **Toxics**, v. 11, n. 2, p. 145, 2023.

LIU, L. *et al*. Characterization and determination of trace alkaloids in seed extracts from *Peganum harmala* Linn. using LC-ESI-MS and HPLC. **Acta Chromatographica**, v. 25, n. 3, p. 555-570, 2013. doi: 10.1556/AChrom.25.2013.3.8.

NENAAH, G. E. Toxicity and growth inhibitory activities of methanol extract and the β-carboline alkaloids of *Peganum harmala* L. against two coleopteran stored-grain pests. **Journal of Stored Products Research**, v. 47, n. 3, p. 255-261, 2011. doi: 10.1016/j.jspr.2011.04.004.

NOORAZLAN, N. A. A. *et al*. Larvicidal potential of plant-based extracts against dengue vector: A short review. **Medical Journal of Malaysia**, v. 79, supl. 1, p. 203-208, mar. 2024. PMID: 38555906.

MORAIS, L. A. S. de; MARINHO-PRADO, J. S**. Plantas com atividade inseticida**. In: *Defensivos agrícolas naturais: uso e perspectivas*. 2016.

MOYES, C. L. *et al*. Contemporary status of insecticide resistance in the major *Aedes* vectors of arboviruses infecting humans. **PLoS Neglected Tropical Diseases**, v. 11, n. 7, p. e0005625, 2017.

PAN AMERICAN HEALTH ORGANIZATION (PAHO). **Dengue in the Americas reaches record in 2023**. Disponível em: https://www.paho.org. Acesso em: 15 abr. 2026.

PASSOS, C. D. *et al*. Alkaloids as inhibitors of monoamine oxidases and their role in the central nervous system. **Studies in Natural Product Chemistry**, v. 43, p. 123-144, 2014.

SANTOS, B. W. L. *et al*. Biodiversity of β-carboline profile of *Banisteriopsis caapi* and ayahuasca, a plant and a brew with neuropharmacological potential. **Plants (Basel)**, v. 9, n. 7, p. 870, 2020. doi: 10.3390/plants9070870. PMID: 32660098; PMCID: PMC7411993.

WORLD HEALTH ORGANIZATION (WHO). **Vector-borne diseases**. Geneva: WHO, 2017. Disponível em: https://www.who.int/news-room/fact-sheets/detail/vector-borne-diseases. Acesso em: 15 abr. 2026.

ZENG, Y. *et al.* Cytotoxic and insecticidal activities of derivatives of harmine, a natural insecticidal component isolated from *Peganum harmala*. **Molecules**, v. 15, n. 11, p. 7775-7791, 2010. doi: 10.3390/molecules15117775.